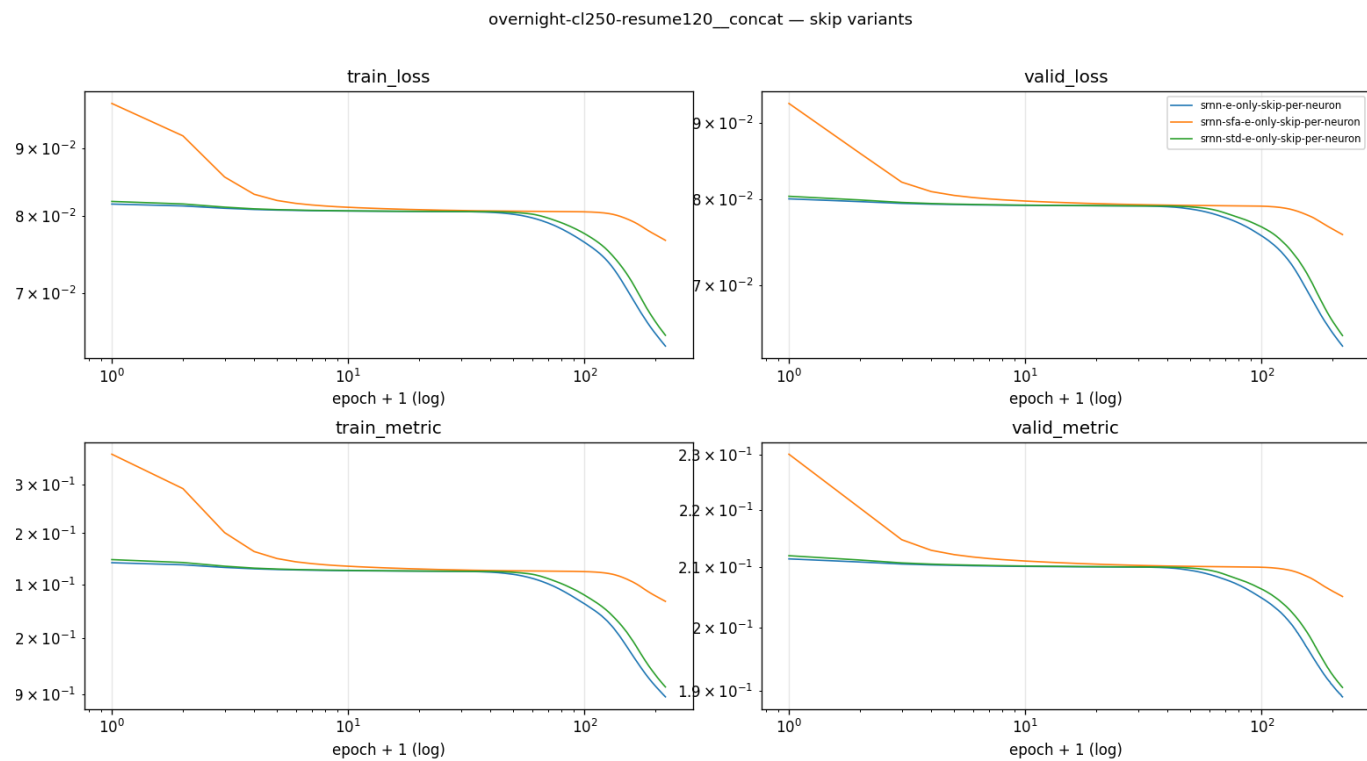


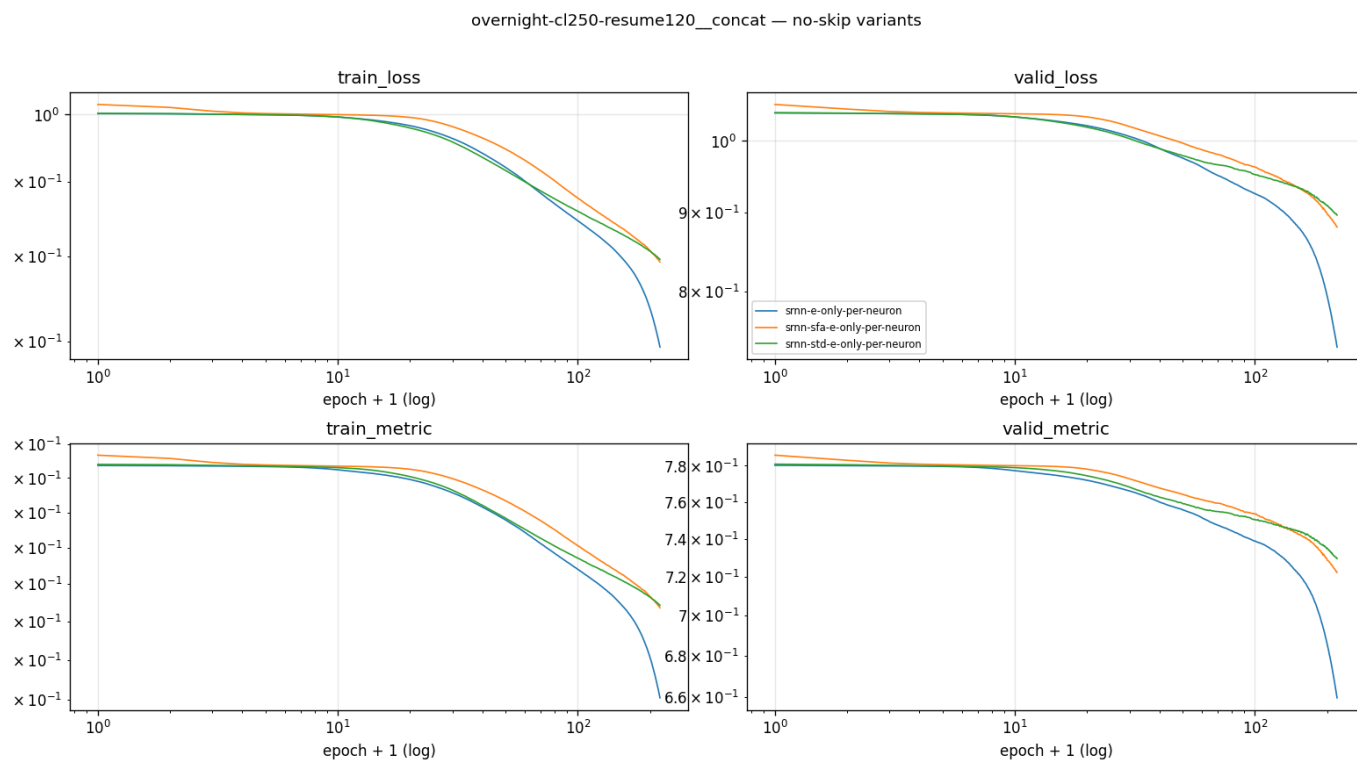
Run: overnight-cl250-resume120__concat

Log-log loss curves



Final test_loss (init - last) [skip]:
 srnn-e-only-skip-per-neuron: 0.07910 → 0.06201
 srnn-sfa-e-only-skip-per-neuron: 0.09419 → 0.07453
 srnn-std-e-only-skip-per-neuron: 0.07961 → 0.06321

Figure 1: skip variants — log-log loss/metric



Final test loss (init - last) [no-skip]:
 srnn-e-only-per-neuron: 0.98188 - 0.66636
 srnn-sfa-e-only-per-neuron: 0.91628 - 0.79321
 srnn-std-e-only-per-neuron: 0.98234 - 0.88718

Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

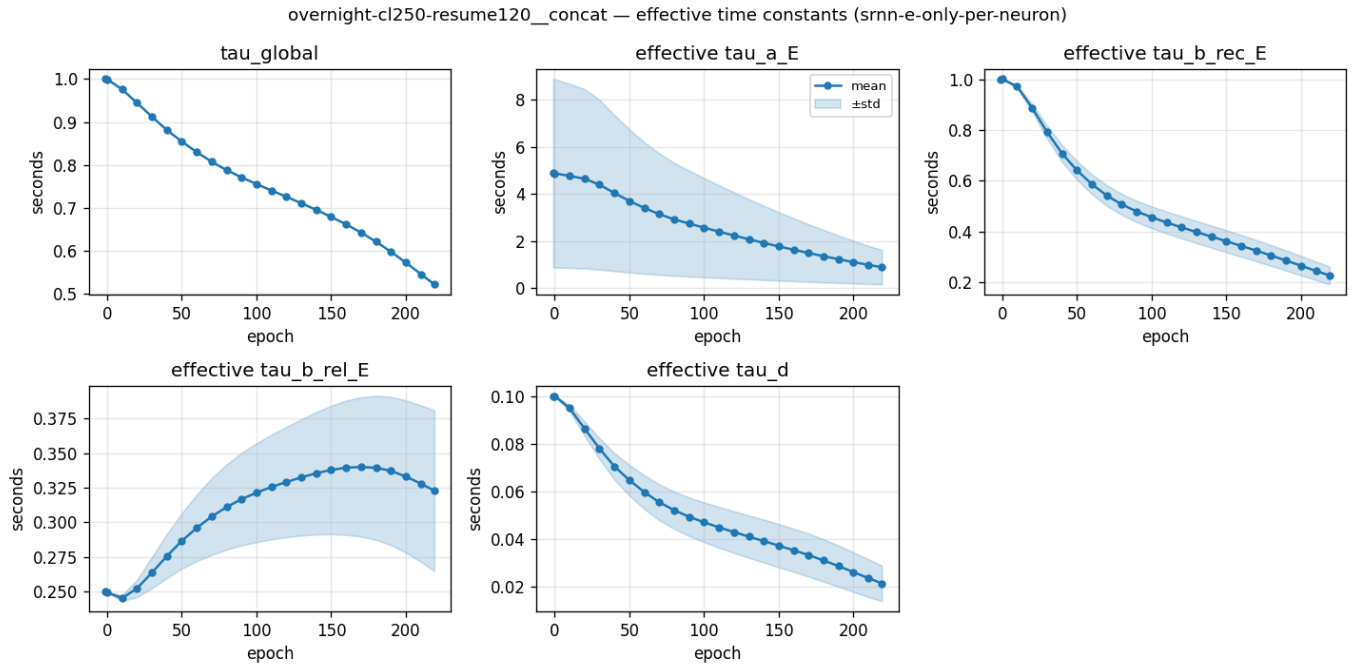
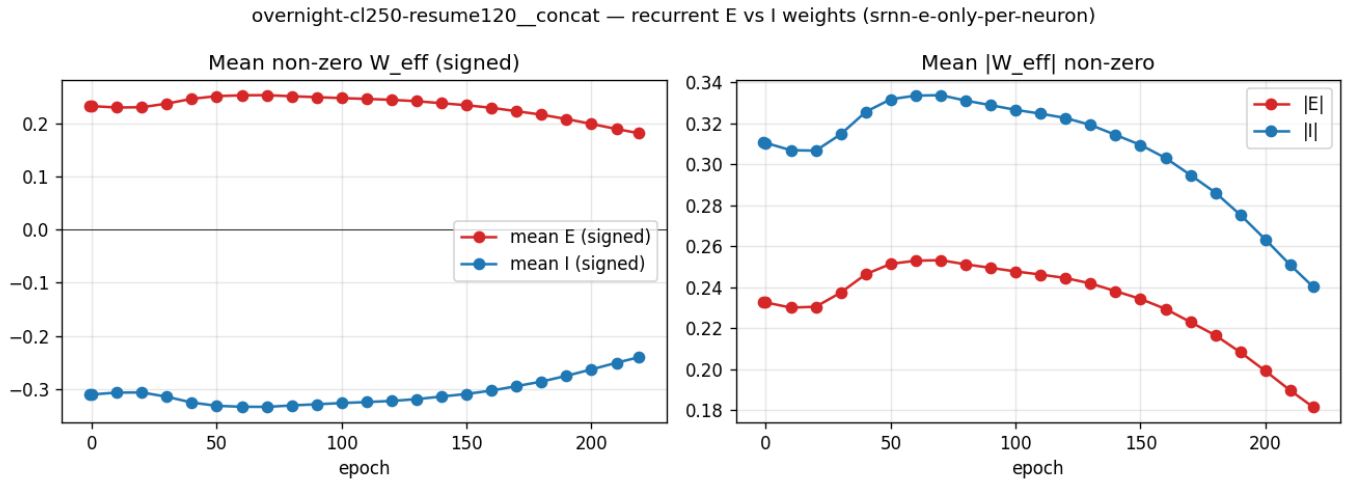


Figure 3: Tau evolution

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

```
# overnight-cl250-resume120_concat - variant srnn-e-only-per-neuron (k=0)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.52182	-	-	scalar
tau_d (s)	+0.1	+0.021479	0	0.007465	300
tau_a_E (s)	+4.8777	+0.89414	3.996	0.7305	450
tau_b_rec_E (s)	+1	+0.22725	0	0.0352	150
tau_b_rel_E (s)	+0.25	+0.32289	0	0.05786	150
W_eff (E src, signed)	+0.23267	+0.18151	0.07641	0.06566	15048
W_eff (I src, signed)	-0.31062	-0.24028	0.07724	0.07004	14977
W_eff (E src)	+0.23267	+0.18151	0.07641	0.06566	15048
W_eff (I src)	+0.31062	+0.24028	0.07724	0.07004	14977
W_in_eff (input neurons)	-0.0003386	+0.00081433	0.1012	0.07879	6675
c_E (SFA coupling)	+0.05	+0.082126	0	0.02424	450
c_0_E (SFA offset)	+0	-0.36987	0	0.1679	450
a_0 (threshold)	+0.35	+0.16904	0	0.1091	300
readout_weight	-2.8854e-05	-0.00058804	0.06604	0.2205	6675
readout_bias	+0	+0.00010319	0	0.2097	89
ic.ic (per-variant init)	+0.098875	-0.055855	0.3842	0.5219	900

srnn-e-only-skip-per-neuron

overnight-cl250-resume120_concat — effective time constants (srnn-e-only-skip-per-neuron)

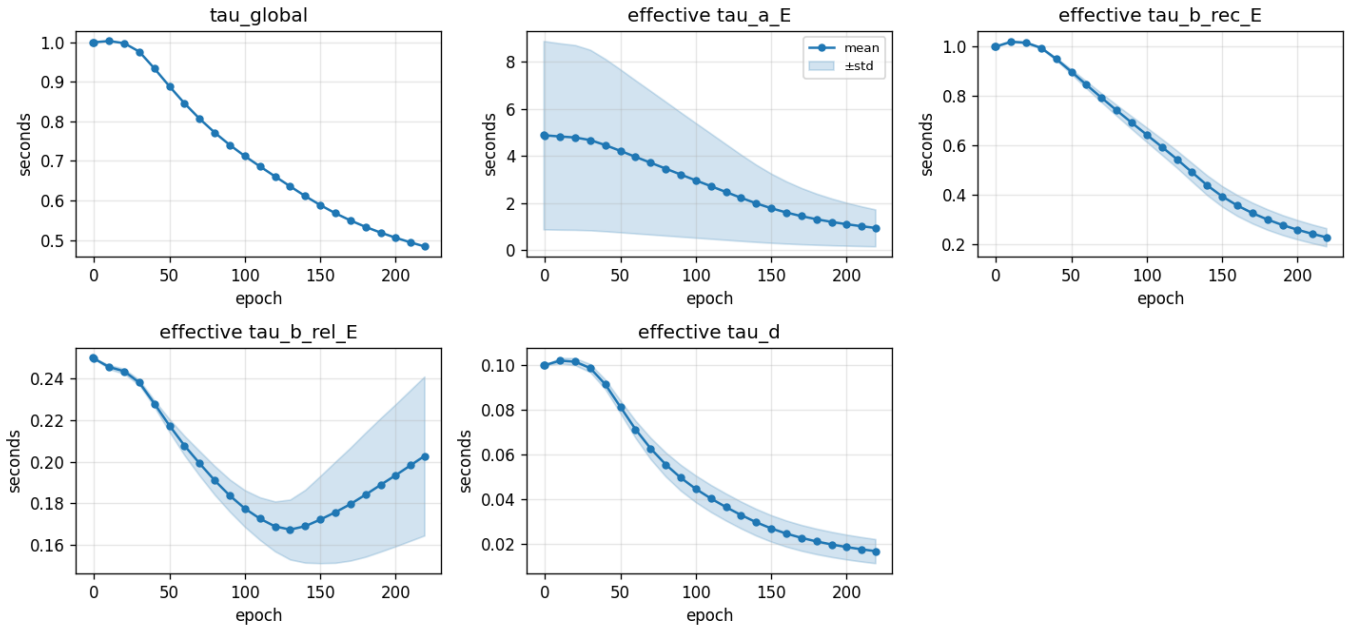
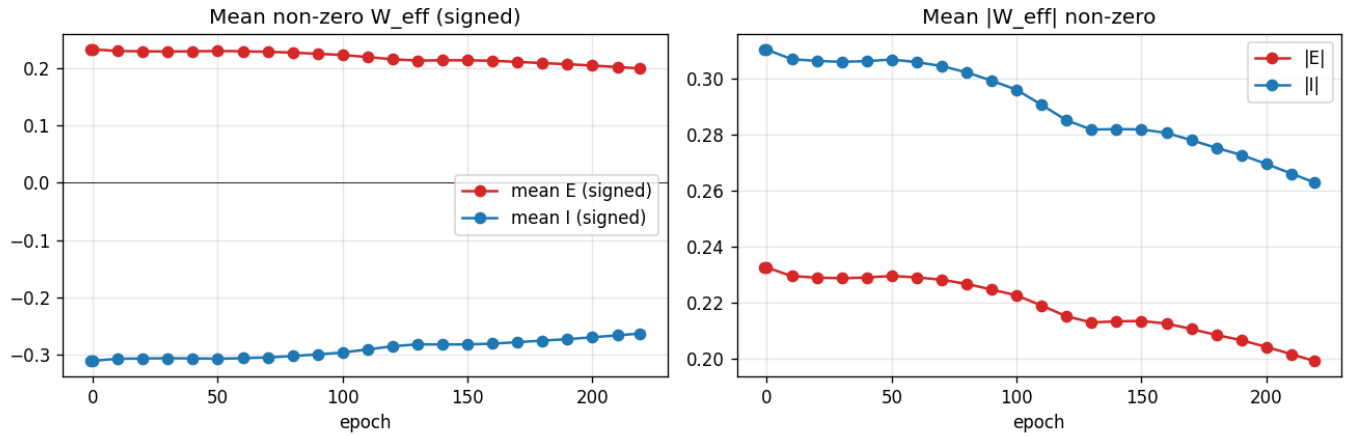


Figure 5: Tau evolution

overnight-cl250-resume120_concat — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6: W_{EI} evolution

srnn-e-only-skip-per-neuron — parameter table

```
# overnight-cl250-resume120_concat - variant srnn-e-only-skip-per-neuron (k=1)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.48399	-	-	scalar
tau_d (s)	+0.1	+0.016801	0	0.005397	300
tau_a_E (s)	+4.8777	+0.94549	3.996	0.7839	450
tau_b_rec_E (s)	+1	+0.22835	0	0.0369	150
tau_b_rel_E (s)	+0.25	+0.20273	0	0.03824	150
W_eff (E src, signed)	+0.23267	+0.19911	0.07641	0.07014	15048
W_eff (I src, signed)	-0.31062	-0.26301	0.07724	0.07369	14977
W_eff (E src)	+0.23267	+0.19911	0.07641	0.07014	15048
W_eff (I src)	+0.31062	+0.26301	0.07724	0.07369	14977
W_in_eff (input neurons)	-0.00084237	-0.00083735	0.1008	0.03921	6675
c_E (SFA coupling)	+0.05	+0.099054	0	0.03138	450
c_0_E (SFA offset)	+0	-0.2607	0	0.1696	450
a_0 (threshold)	+0.35	+0.14095	0	0.1374	300
readout_weight	+0.00078219	-0.0052303	0.06654	0.135	6675
readout_bias	+0	+0.11501	0	0.1561	89
ic.ic (per-variant init)	+0.098961	-0.032454	0.387	0.5247	900

srnn-sfa-e-only-per-neuron

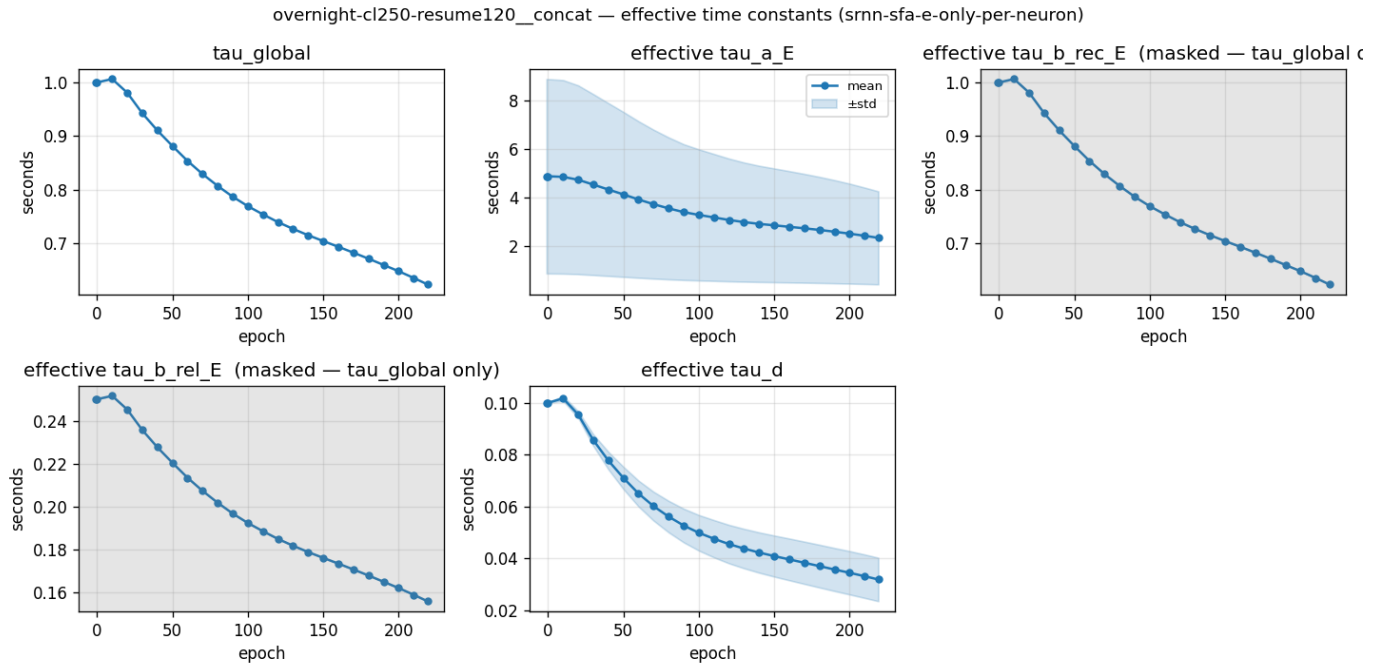
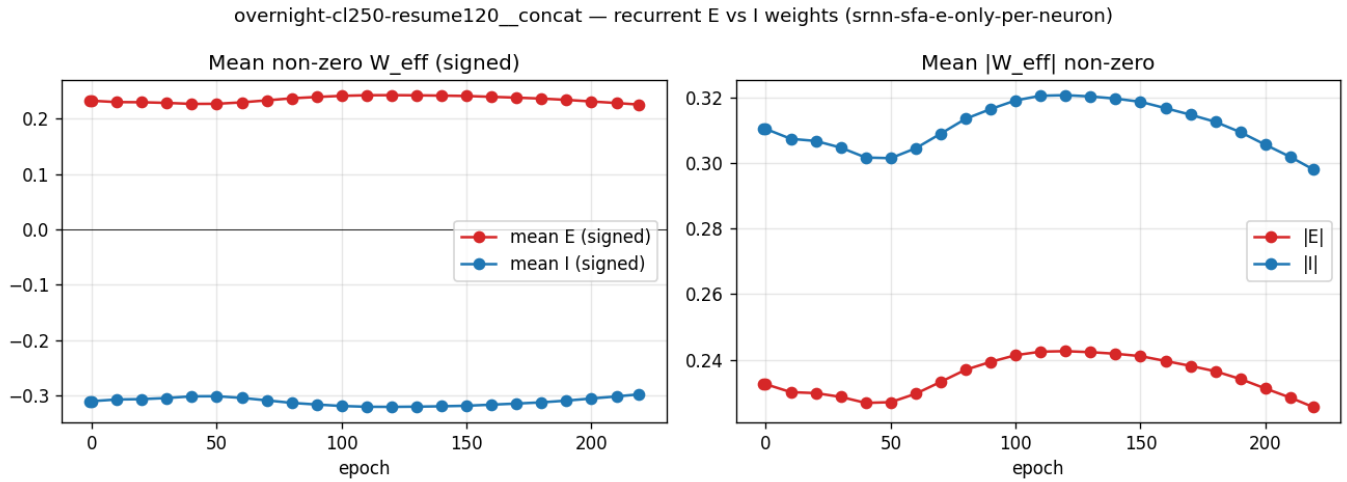


Figure 7: Tau evolution

Figure 8: W_{EI} evolution

srnn-sfa-e-only-per-neuron — parameter table

```
# overnight-cl250-resume120_concat - variant srnn-sfa-e-only-per-neuron (k=2)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.6228	-	-	scalar
tau_d (s)	+0.1	+0.031946	0	0.008385	300
tau_a_E (s)	+4.8777	+2.3462	3.996	1.912	450
tau_b_rec_E (s)	+1	+0.6228	0	0	150
tau_b_rel_E (s)	+0.25	+0.1557	0	0	150
W_eff (E src, signed)	+0.23267	+0.22561	0.07641	0.07892	15048
W_eff (I src, signed)	-0.31062	-0.29814	0.07724	0.08305	14977
W_eff (E src)	+0.23267	+0.22561	0.07641	0.07892	15048
W_eff (I src)	+0.31062	+0.29814	0.07724	0.08305	14977
W_in_eff (input neurons)	-0.00015382	-0.0018244	0.09934	0.1217	6675
c_E (SFA coupling)	+0.05	+0.045337	0	0.005954	450
c_0_E (SFA offset)	+0	-0.21542	0	0.1249	450
a_0 (threshold)	+0.35	+0.366	0	0.08314	300
readout_weight	-0.00098972	+0.0084713	0.06678	0.1704	6675
readout_bias	+0	-0.10435	0	0.1289	89
ic.ic (per-variant init)	+0.13078	-0.0045598	0.4172	0.4608	900

srnn-sfa-e-only-skip-per-neuron

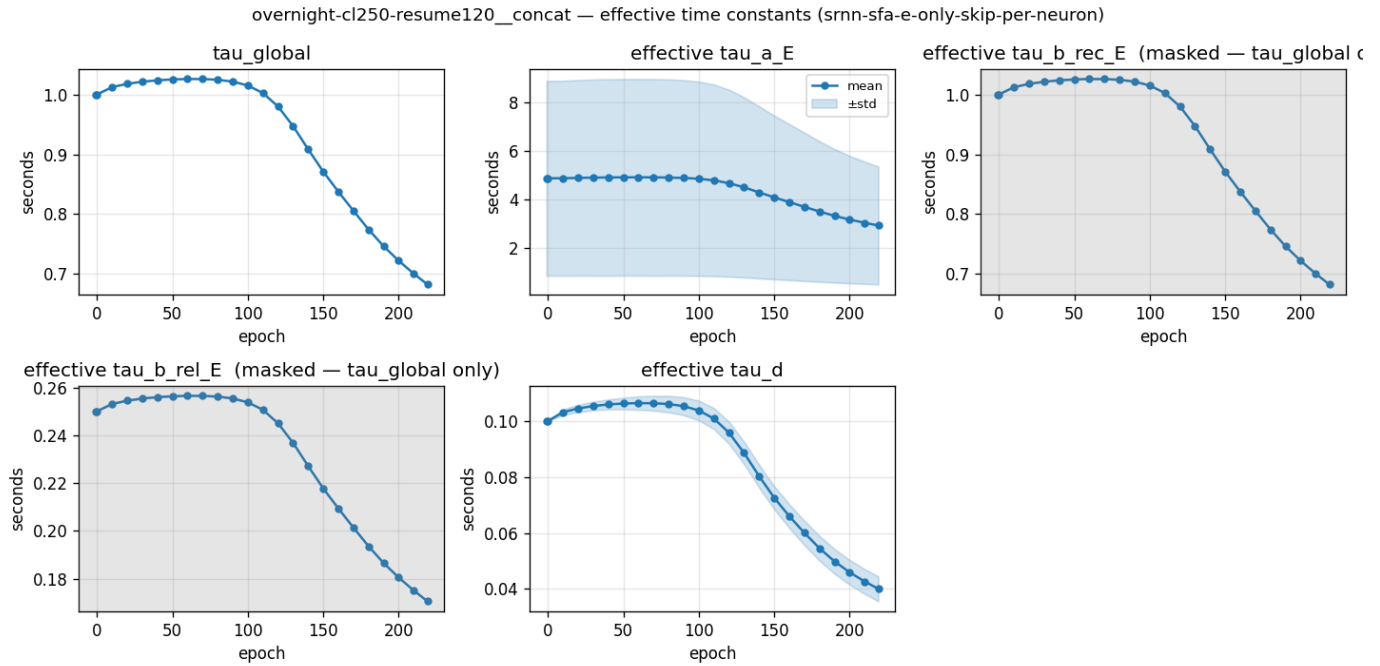


Figure 9: Tau evolution

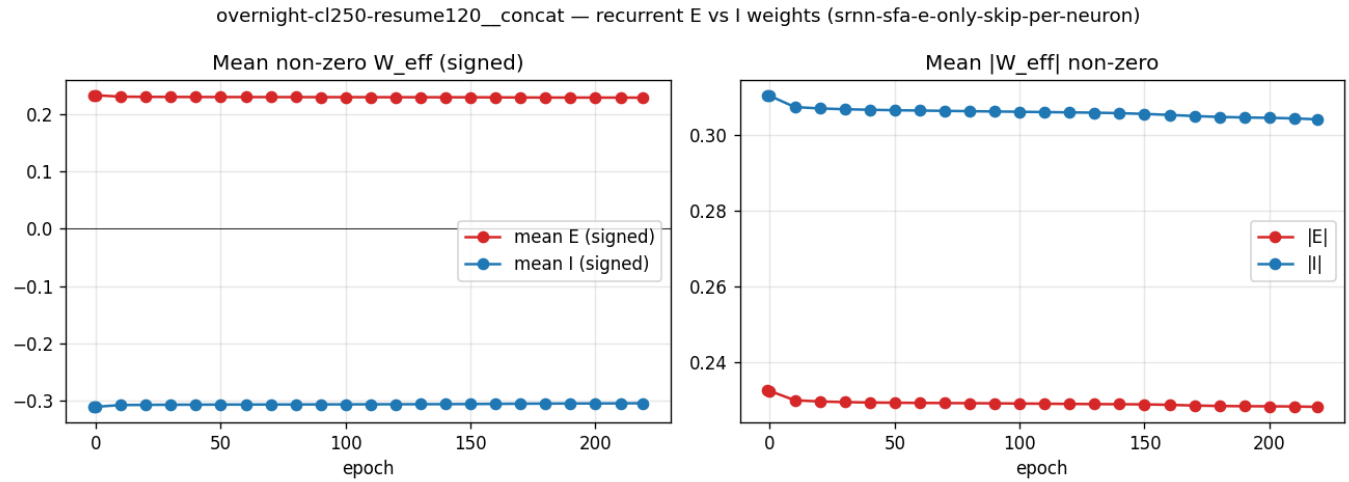


Figure 10: W_EI evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

```
# overnight-cl250-resume120_concat - variant srnn-sfa-e-only-skip-per-neuron (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.68228	-	-	scalar
tau_d (s)	+0.1	+0.040003	0	0.004433	300
tau_a_E (s)	+4.8777	+2.9413	3.996	2.417	450
tau_b_rec_E (s)	+1	+0.68228	0	0	150
tau_b_rel_E (s)	+0.25	+0.17057	0	0	150
W_eff (E src, signed)	+0.23267	+0.22832	0.07641	0.07621	15048
W_eff (I src, signed)	-0.31062	-0.30425	0.07724	0.07747	14977
W_eff (E src)	+0.23267	+0.22832	0.07641	0.07621	15048
W_eff (I src)	+0.31062	+0.30425	0.07724	0.07747	14977
W_in_eff (input neurons)	-0.0011621	-0.0030177	0.09962	0.07681	6675
c_E (SFA coupling)	+0.05	+0.053708	0	0.007939	450
c_0_E (SFA offset)	+0	+0.017061	0	0.1012	450
a_0 (threshold)	+0.35	+0.35718	0	0.0842	300
readout_weight	+0.00016801	-0.0028867	0.0673	0.06855	6675
readout_bias	+0	+0.0371	0	0.04487	89
ic.ic (per-variant init)	+0.09231	+0.054494	0.5294	0.4889	900

srnn-std-e-only-per-neuron

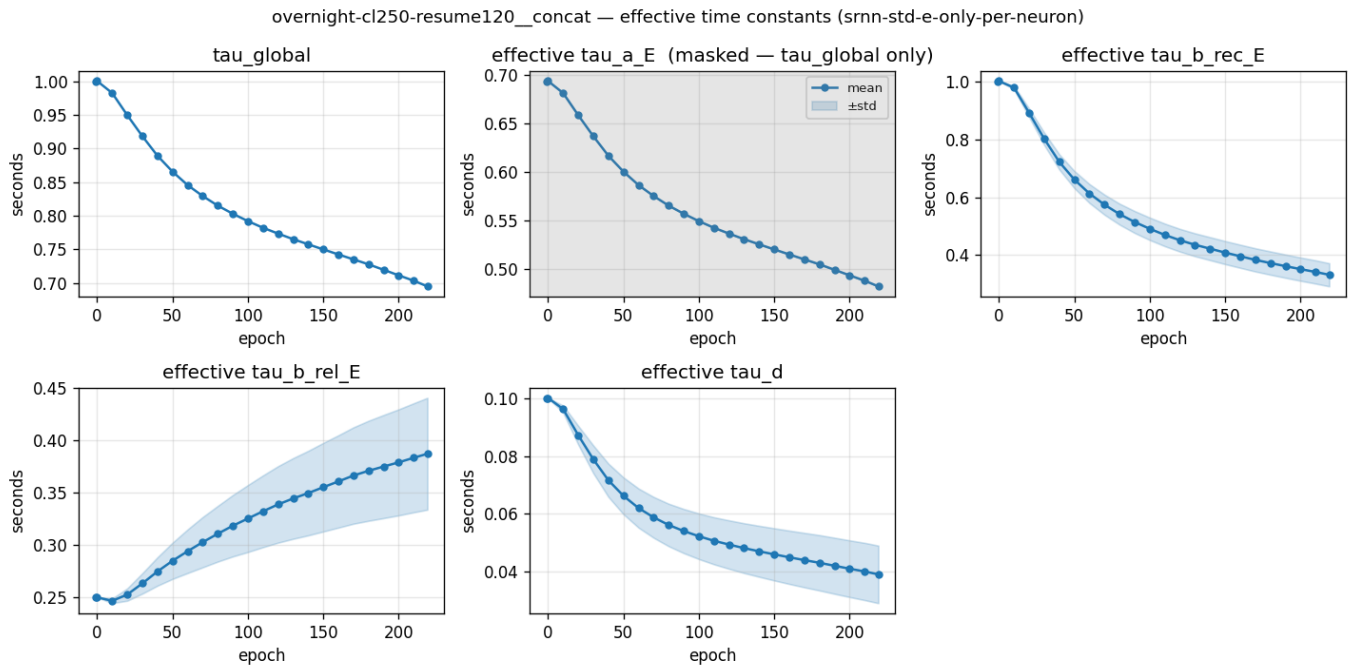
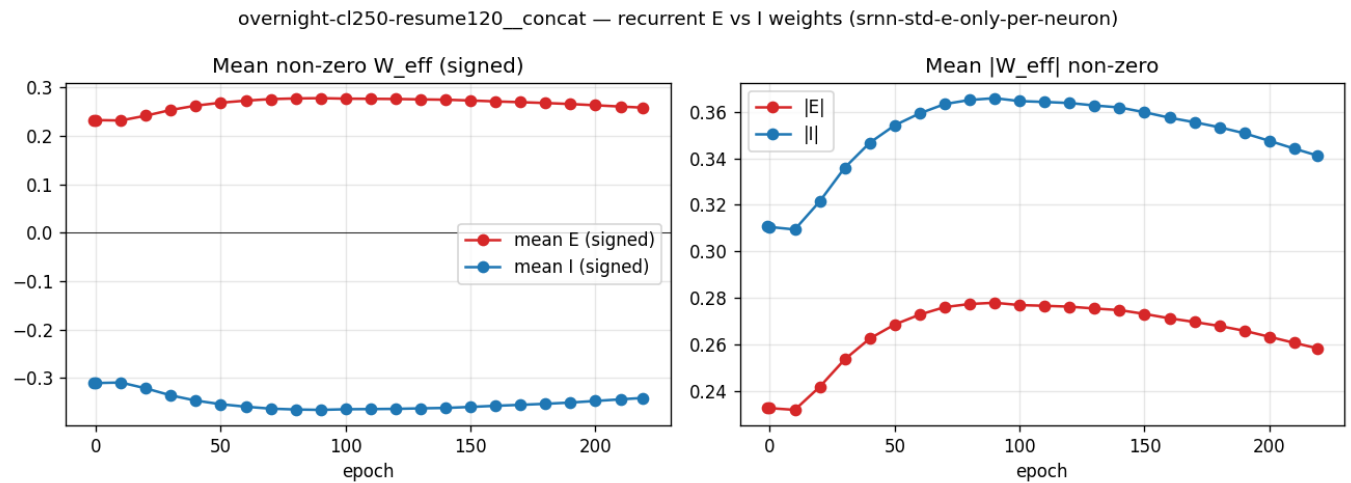


Figure 11: Tau evolution

Figure 12: W_{EI} evolution

srnn-std-e-only-per-neuron — parameter table

overnight-cl250-resume120_concat - variant srnn-std-e-only-per-neuron (k=4)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.69512	-	-	scalar
tau_d (s)	+0.1	+0.039022	0	0.009972	300
tau_a_E (s)	+0.69315	+0.48182	0	0	450
tau_b_rec_E (s)	+1	+0.3318	0	0.03998	150
tau_b_rel_E (s)	+0.25	+0.387	0	0.05346	150
W_eff (E src, signed)	+0.23267	+0.25831	0.07641	0.09257	15048
W_eff (I src, signed)	-0.31062	-0.34123	0.07724	0.09821	14977
W_eff (E src)	+0.23267	+0.25831	0.07641	0.09257	15048
W_eff (I src)	+0.31062	+0.34123	0.07724	0.09821	14977
W_in_eff (input neurons)	+0.00019944	-0.00098562	0.09994	0.1332	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.28106	0	0.1036	300
readout_weight	+0.00074661	+0.010267	0.06657	0.1858	6675
readout_bias	+0	-0.022221	0	0.187	89
ic.ic (per-variant init)	+0.087739	+0.030821	0.3868	0.4926	900

srnn-std-e-only-skip-per-neuron

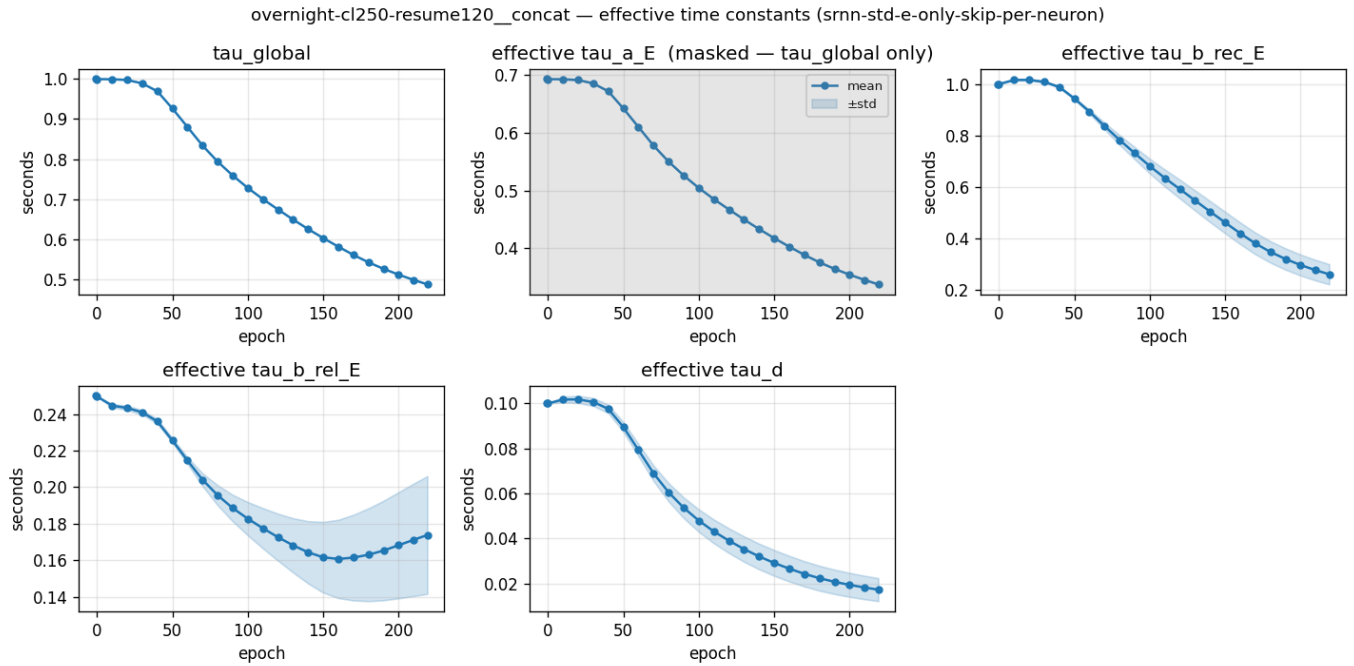
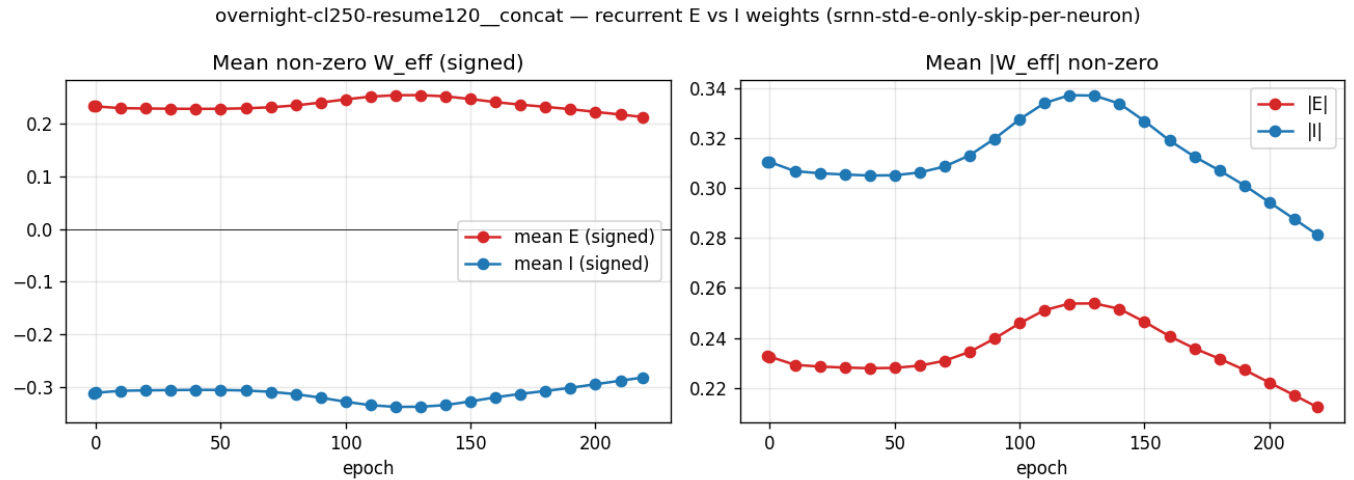


Figure 13: Tau evolution

Figure 14: W_{EI} evolution

srnn-std-e-only-skip-per-neuron — parameter table

```
# overnight-cl250-resume120_concat - variant srnn-std-e-only-skip-per-neuron (k=5)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.48709	-	-	scalar
tau_d (s)	+0.1	+0.01728	0	0.005127	300
tau_a_E (s)	+0.69315	+0.33763	0	0	450
tau_b_rec_E (s)	+1	+0.25985	0	0.03932	150
tau_b_rel_E (s)	+0.25	+0.17382	0	0.03227	150
W_eff (E src, signed)	+0.23267	+0.2124	0.07641	0.07436	15048
W_eff (I src, signed)	-0.31062	-0.28145	0.07724	0.07743	14977
W_eff (E src)	+0.23267	+0.2124	0.07641	0.07436	15048
W_eff (I src)	+0.31062	+0.28145	0.07724	0.07743	14977
W_in_eff (input neurons)	+0.00036119	+7.5388e-06	0.09956	0.03832	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.20351	0	0.1406	300
readout_weight	+0.00055372	-0.0079787	0.06572	0.1213	6675
readout_bias	+0	+0.050723	0	0.1257	89
ic.ic (per-variant init)	+0.089141	-0.00010804	0.3857	0.4971	900